sys.columns (Transact-SQL)

Applies to: SQL Server · Azure SQL Database · Azure SQL Managed Instance · Azure Synapse Analytics · Analytics Platform System (PDW) · SQL analytics endpoint in Microsoft Fabric · Warehouse in Microsoft Fabric · SQL database in Microsoft Fabric

Returns a row for each column of an object that has columns, such as views or tables. The following list contains the object types that have columns:

- Table-valued assembly functions (FT)
- Inline table-valued SQL functions (IF)
- Internal tables (IT)
- System tables (S)
- Table-valued SQL functions (TF)
- User tables (U)
- Views (V)

Expand table

Column name	Data type	Description
`object_id`	**int**	ID of the object to which this column belongs.
`name`	**sysname**	Name of the column. Is unique within the object.
`column_id`	**int**	ID of the column. Is unique within the object. Column IDs might not be sequential.
`system_type_id`	**tinyint**	ID of the system type of the column.
`user_type_id`	**int**	ID of the type of the column as defined by the user. To return the name of the type, join to the sys.types catalog view on this column.
`max_length`	**smallint**	Maximum length (in bytes) of the column. `-1` = Column data type is **varchar(max)**, **nvarchar(max)**, **varbinary(max)**, or **xml**. For **text**, **ntext**, and **image** columns, the `max_length` value is `16` (representing the 16-byte pointer only) or the value set by `sp_tableoption 'text in row'`.

Column name	Data type	Description
`precision`	**tinyint**	Precision of the column if numeric-based; otherwise, `0`.
`scale`	**tinyint**	Scale of column if numeric-based; otherwise, `0`.
`collation_name`	**sysname**	Name of the collation of the column if character-based; otherwise `NULL`.
`is_nullable`	**bit**	`1` = Column is nullable `0` = Column isn't nullable
`is_ansi_padded`	**bit**	`1` = Column uses `ANSI_PADDING ON` behavior if character, binary, or variant `0` = Column isn't character, binary, or variant
`is_rowguidcol`	**bit**	`1` = Column is a declared `ROWGUIDCOL`
`is_identity`	**bit**	`1` = Column has identity values
`is_computed`	**bit**	`1` = Column is a computed column
`is_filestream`	**bit**	`1` = Column is a FILESTREAM column
`is_replicated`	**bit**	`1` = Column is replicated
`is_non_sql_subscribed`	**bit**	`1` = Column has a non-SQL Server subscriber
`is_merge_published`	**bit**	`1` = Column is merge-published
`is_dts_replicated`	**bit**	`1` = Column is replicated by using SSIS
`is_xml_document`	**bit**	`1` = Content is a complete XML document `0` = Content is a document fragment, or the column data type isn't **xml**
`xml_collection_id`	**int**	Nonzero if the data type of the column is **xml** and the XML is typed. The value is the ID of the collection containing the validating XML schema namespace of the column `0` = No XML schema collection
`default_object_id`	**int**	ID of the default object, regardless of whether it's a stand-alone object sp_bindefault, or an inline, column-level `DEFAULT` constraint. The `parent_object_id` column of an inline column-level default object is a reference back to the table itself.

Column name	Data type	Description
		`0` = No default
`rule_object_id`	**int**	ID of the stand-alone rule bound to the column by using `sys.sp_bindrule`. `0` = No stand-alone rule. For column-level `CHECK` constraints, see sys.check_constraints.
`is_sparse`	**bit**	`1` = Column is a sparse column. For more information, see Use sparse columns.
`is_column_set`	**bit**	`1` = Column is a column set. For more information, see Use sparse columns.

`generated_always_type`	**tinyint**	Identifies when the column value is generated (is always `0` for columns in system tables).
		
		Applies to: SQL Server 2016 (13.x) and later versions, and SQL Database.
		
		`0` = `NOT_APPLICABLE`
		`1` = `AS_ROW_START`
		`2` = `AS_ROW_END`
		
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database.
		
		`5` = `AS_TRANSACTION_ID_START`
		`6` = `AS_TRANSACTION_ID_END`
		`7` = `AS_SEQUENCE_NUMBER_START`
		`8` = `AS_SEQUENCE_NUMBER_END`
		
		For more information, see Temporal tables.

`generated_always_type_desc`	**nvarchar(60)**	Textual description of the `generated_always_type` value (always `NOT_APPLICABLE` for columns in system tables)
		
		Applies to: SQL Server 2016 (13.x) and later versions, and SQL Database.
		
		`NOT_APPLICABLE`
		`AS_ROW_START`
		`AS_ROW_END`
		
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database.

Column name	Data type	Description

		`AS_TRANSACTION_ID_START`
		`AS_TRANSACTION_ID_END`
		`AS_SEQUENCE_NUMBER_START`
		`AS_SEQUENCE_NUMBER_END`

`encryption_type`	**int**	Encryption type: `1` = Deterministic encryption `2` = Randomized encryption **Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database

`encryption_type_desc`	**nvarchar(64)**	Encryption type description:
		
		`RANDOMIZED`
		`DETERMINISTIC`
		
		Applies to: SQL Server 2016 (13.x) and later versions, and SQL Database

`encryption_algorithm_name`	**sysname**	Name of encryption algorithm. Only `AEAD_AES_256_CBC_HMAC_SHA_512` is supported. **Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database
`column_encryption_key_id`	**int**	ID of the column encryption key (CEK). **Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database
`column_encryption_key_database_name`	**sysname**	The name of the database where the column encryption key exists if different than the database of the column. `NULL` if the key exists in the same database as the column. **Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database
`is_hidden`	**bit**	Indicates if the column is hidden: `0` = regular, not-hidden, visible column `1` = hidden column **Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database

Column name	Data type	Description
`is_masked`	**bit**	Indicates if the column is masked by dynamic data masking: `0` = regular, not-masked column `1` = column is masked **Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database
`graph_type`	**int**	Internal column with a set of values. The values are between `1` and `8` for graph columns, and `NULL` for others.
`graph_type_desc`	**nvarchar(60)**	Identified for informational purposes only. Not supported. Future compatibility is not guaranteed.
`is_data_deletion_filter_column`	**bit**	Indicates if the column is the data retention filter column for the table. **Applies to**: Azure SQL Edge

`ledger_view_column_type`	**int**	If not `NULL`, indicates the type of a column in a ledger view:
		
		`1` = `TRANSACTION_ID`
		`2` = `SEQUENCE_NUMBER`
		`3` = `OPERATION_TYPE`
		`4` = `OPERATION_TYPE_DESC`
		
		For more information, see Ledger overview.
		
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database

`ledger_view_column_type_desc`	**nvarchar(60)**	If not `NULL`, contains a textual description of the type of a column in a ledger view:
		
		`TRANSACTION_ID`
		`SEQUENCE_NUMBER`
		`OPERATION_TYPE`
		`OPERATION_TYPE_DESC`
		
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database

`is_dropped_ledger_column`	**bit**	Indicates a ledger table column that was dropped.

Column name	Data type	Description
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database
`vector_dimensions`	**int**	Indicates how many dimensions the vector has. **Applies to**: SQL Server 2025 (17.x) and later versions, and SQL Database
`vector_base_type`	**tinyint**	Indicates the data type used to store vector dimensions values. `0` = 32-bit (single-precision) float `1` = 16-bit (half-precision) float [1] **Applies to**: SQL Server 2025 (17.x) and later versions, and SQL Database
`vector_base_type_desc`	**nvarchar(10)**	Contains the textual description of the data type used to store vector dimensions values. **Applies to**: SQL Server 2025 (17.x) and later versions, and SQL Database

[1] For more information, see Half-precision floating-point format.

Permissions

The visibility of the metadata in catalog views is limited to securables that a user either owns, or on which the user was granted some permission. For more information, see Metadata visibility configuration.

Usage examples

Get column details for a table

To get metadata for columns in a table, use the following code:

SQL

```
CREATE TABLE dbo.[sample]
(
    id INT NOT NULL,
    col1 VARBINARY (10) NULL
);
```

```
SELECT c.[name] AS column_name,
       t.[name] AS [type_name],
       c.[max_length],
       c.[precision],
       c.[scale]
FROM sys.columns AS c
     INNER JOIN sys.types AS t
         ON c.user_type_id = t.user_type_id
WHERE object_id = object_id('dbo.sample');
```

Related content

- Object catalog views (Transact-SQL)
- System catalog views (Transact-SQL)
- Querying the SQL Server System Catalog FAQ
- sys.all_columns (Transact-SQL)
- sys.system_columns (Transact-SQL)
- sys.types (Transact-SQL)

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